Girls Build Bots — Junior Cybersecurity Basics

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Cybersecurity — Cybersecurity Basics **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools and simple props.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: being safe and smart online

The series builds real digital literacy: understand how information travels (**Basics**), protect it (Safeguards), and investigate critically (Investigator). Juniors can handle the how-it-works layer — networks, messages hopping between computers, and why data is valuable.

Vocabulary: network, data, message, router, personal information, digital footprint, encrypt (secret code).

Cybersecurity Basics (90–120 minutes)

Badge theme: How information travels across networks, what makes data valuable — and vulnerable.

Materials

- Index cards + envelopes for the human-network game; one “router” sign per group
- A simple substitution cipher sheet (A=1, B=2… or a Caesar wheel printout)
- Devices with **Interland** (beinternetawesome.withgoogle.com) or CYBER.ORG activities
- “Data value” cards (your name, your school, your password, your favorite song, your address)

Activity 1 — The Human Internet (25 min)

Build a network out of Scouts. Senders write a short message on a card, seal it in an envelope addressed to a receiver across the room — but it must travel **hand to hand through “routers”** (Scouts in the middle) to get there. Round 2: a playful “snoop” router peeks at

unsealed cards. **Debrief:** on the internet, messages hop through many computers; anyone in the path could look — which is why some messages need protecting.

Activity 2 — Secret Codes (Encryption) (20 min)

Solution to the snoop: write the message in a **cipher** before sending. Pairs encode and decode short messages with the substitution sheet. The snoop gets a coded card and can't read it. That's the core idea of **encryption** — the internet's envelope.

Activity 3 — What's Your Data Worth? (20 min)

Sort the data-value cards: which would a stranger most want, and why? (Password > address > school > favorite song.) Introduce **digital footprint**: the trail of information you leave online adds up. Juniors are ready for the honest version: data is valuable, so people try to collect it — which is why we choose what to share.

Activity 4 — Interland Round (20 min) + Wrap (10 min)

Play a realm reinforcing share-with-care. Closing circle: each Scout names one thing she learned about how information travels, and one piece of data she'll guard.

> **Robot moment (optional):** Share for 60 seconds how the team's robot and driver station talk over a network during matches — and why the field's network is locked down so no one else can send the robot commands.

Completion Checklist — Cybersecurity Basics

By the end of the session, to earn the badge, each Scout can: - [ ] Explain how a message travels through a network of computers - [ ] Explain why some information needs protecting on its way - [ ] Encode and decode a message with a simple cipher - [ ] Rank types of personal data by value and explain her reasoning

---

Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Envelopes, ciphers, and a browser game carry the session. The "robot moment" is a 60-second add-on.
- **Make the snoop playful, not creepy.** A cartoonish nosy router lands the lesson without fear.
- **Let the cipher click.** The moment the snoop can't read the coded card is the whole point — stage it deliberately.
- **Honest but calm on data value.** Juniors can handle "your data is valuable"; keep the tone matter-of-fact, never alarming.
- **Connect every game moment back** to the human-network activity — the physical memory anchors the concept.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.